ĐẶNG BẢO PHONG

DevOps Engineer

0937375404 michaeldang.general@gmail.com Bien Hoa, Dong Nai

OBJECTIVE

DevOps Engineer with strong coding, automation, and systems skills, looking to design and operate scalable, secure infrastructure while mentoring teams—progressing toward **Solutions Architect** responsibilities in end-to-end solution design.

EDUCATION

University of Science Center Feb 2018 - Aug 2018

Python programming

- Basic & Advanced Python programming
- Web application deployment with Django

DONG NAI TECHNICAL COLLEGE Aug 2016 - Feb 2018

INFORMATION TECHNOLOGY

WORK EXPERIENCE

Home Credit Viet Nam Oct 2024 - Present

DevOps Engineer

- Engineered and implemented a **middleware** solution to seamlessly integrate **Genesys Chat** with **Zalo OA**. This project enabled a new, direct communication channel, improving customer reach and service efficiency.
- Implemented **GitOps** workflows to streamline and enhance the deployment process for **Genesys Call Center** systems, ensuring efficient and reliable operations.
- Research and engineering of **CallCraft**, an automation service designed for the Genesys Call Center System using the Genesys SDK platform.
- Designed and built **CCSS**, a self-service operations portal for contact-center teams, using React + TypeScript and Python microservices with PostgreSQL — enabling **business users** to manage key tasks without IT involvement and significantly improving operational throughput.
- Designed, built, and integrated **GitLab CI/CD** pipelines, automating deployment processes to improve productivity and reduce errors.
- Researched and optimized **containerization** of **Genesys components** and applications, facilitating a more scalable and portable architecture.

INCEPTIONLABS May 2022 - Oct 2024

Operation Engineer

- Integrated GitLab CI/CD with Kubernetes clusters, enabling automatic deployment to development, staging, and production environments
- Created and configured ECR repositories to host Docker images for various microservices and applications deployed on AWS infrastructure
- **Orchestrated microservices on Amazon EKS** (Deployments/Services/Ingress): improved **node/pod utilization +20%**, achieved **99.95%** service availability.
- Designed and implemented Amazon EFS file systems to provide persistent volume for the Kubernetes cluster
- **Performance troubleshooting with dev teams:** optimized limits/requests and GC/probes → **CPU -25%**, **memory -17%**, **error rate -30%** across top paths.
- **Multi-cluster ready** (if applicable): isolated Argo CD Projects per team/app domain; standard release policies and sync windows across clusters; onboarding new services **1 day → 2 hours**.
- **Automated configuration with Ansible:** idempotent roles for app/runtime/OS hardening → manual errors **-90%**, deployment time **-70%**

- **Automated EKS provisioning & autoscaling** via Terraform/Ansible: cut **scale-up time** from **30 min → 7 min (-76%)**; saved **~12%** monthly through rightsizing & Spot usage.
- Implemented monitoring stack PGAK to keep track of system resources; using Grafana cloud for Kubernetes cluster monitoring
- **Centralized logging with Fluent Bit + Loki + Grafana:** aggregated logs for **100+** workloads.
- Implemented Rancher Fleet for managing and deploying Kubernetes configurations across multiple clusters, enhancing consistency and governance
- Implemented Cloudflare's page rules and worker scripting for customizing and optimizing website behavior
- **Python deployment tooling:** automated server/app provisioning (infra + app stacks), reduced human steps **-90%**.
- **Operational documentation & runbooks:** playbooks and troubleshooting guides.

PLS Integrated Technology Services JSC Dec 2018 - Jan 2023

Sysadmin/Ops

- **Replaced Asterisk with an OpenSIPS-based SIP proxy**, optimizing compute/memory usage and handling high call volumes reliably.
- **Designed RESTful APIs (Python/Flask)** to serve OpenSIPS custom operations (e.g., routing, dial-plan checks), enabling seamless system-to-system communication.
- **Implemented observability for OpenSIPS** with **Prometheus, Grafana, and HOMER**, plus targeted Python probes—tracking performance, statistics, and QoS with actionable alerts.
- **Standardized local/prod parity with Docker Compose** → dev env spin-up **1 hour → 5 min (-91%).**
- Implemented dynamic configuration scripts in Jenkins to adjust SIP proxy settings based on the deployment environment
- Led the setup and ongoing maintenance of the call center infrastructure, ensuring seamless operation of telephony systems for some major customers, i.e., VPBank, Sacobe, FEcredit,...
- **Coordinated interoperability with vendors and third-party SIP carriers**, validating signaling and media flows for smooth cross-network communication.
- **Debian server provisioning + Ansible** → build/config **40 min → 8 min (-80%)**; config drift **6/mo → 0.**
- Engineered headless browser automation using Selenium in Python for scenarios where GUI interactions were not necessary, optimizing script execution in server environments
- Developed custom Odoo modules to extend existing functionalities and meet specific business needs
- Offered on-site technical assistance to call center agents, resolving hardware and software issues promptly to minimize disruptions

CERTIFICATES

Python Django Web development basic	**2018**
Python Django Web development advance	**2018**

HONORS & AWARDS

Best performance of year of PLS	**2021**

SKILLS

Core Tech Stack	Linux; Python; Go; Bash; Node.js; Django; Docker; Docker Swarm; Kubernetes; Rancher; GitLab CI/CD; Jenkins; Terraform; AWS; Prometheus; Grafana; Fluent Bit; Loki; NGINX; Apache; PostgreSQL; MySQL; NoSQL; Asterisk; OpenSIPS/Kamailio; Cloudflare; Selenium; REST API.
Communication Skills	• Active listening • Written communication
Team work	• Collaboration
Problem-Solving	• Decision-making
Adaptability	• Flexibility • Openness to change • Ability to learn quickly

PROJECTS

CallCraft Jan 2025 - Present

Position: Dev/DevOps

Customer: IT Department
Team size: 3
Tech stack: React + TypeScript, Java Spring Boot, FastAPI, Keycloak for auth.

- **Orchestrated the full product lifecycle** of "CallCraft," a custom Genesys automation platform, from architectural design and full-stack development to containerization and production deployment.
- **Designed a hybrid microservices architecture** leveraging **Java Spring Boot** for transactional stability and **FastAPI** for high-performance automation, integrated directly with the **Genesys SDK**.
- **Developed a type-safe, responsive frontend** using **React** and **TypeScript**, ensuring seamless state management between the custom UI and backend Genesys events.
- **Implemented DevSecOps best practices** by configuring **Keycloak** for centralized OIDC authentication and managing secure token exchanges across distributed services.

CCSS Portal May 2025 - Present

Position: Dev/DevOps

Customer: Business Department
Team size: 2
Tech stack: React + TypeScript, Node.js, Python, PostgreSQL database, Keycloak for auth, F5 for routing/load balancing.

- Designed and implemented the **full-stack CallCraft Portal**, building the frontend in TypeScript (React) and backend services in **Node.js** and **Python** with **PostgreSQL** as the primary data store
- Integrated **Keycloak** for authentication and SSO, including role-based access control so business users can safely perform self-service operations.
- Worked with **F5** for routing and reverse proxy configuration, ensuring secure, reliable access to the portal across environments.
- Owned end-to-end delivery: architecture, implementation, deployment, and ongoing enhancements based on feedback from contact-center and business users.

Zalo-Genesys Chat Integration March 2025 - Dec 2025

Position: Dev/DevOps

Customer: Business Department
Team size: 3
Tech stack: Docker, Genesys Components, Nodejs, Redis

- **Zalo-Genesys Integration:** Architected and built a robust Node.js middleware solution to serve as a webhook service, effectively bridging the communication gap between the Zalo platform and the Genesys Engage contact center.
- **Webhook Development:** Engineered the primary webhook service to securely receive, process, and relay real-time chat messages and events between Zalo users and Genesys agents.
- **Containerized eService Deployment:** Packaged and deployed the Genesys eService components using **container technology (e.g., Docker)**. This approach ensured a consistent, isolated, and scalable environment for the chat workflow engine.
- **Genesys eService Implementation:** Within the containerized instance, designed and implemented chat workflows to handle incoming Zalo interactions, ensuring proper routing, queuing, and agent assignment.

Migrate traditional services to Kubernetes July 2023 - Oct 2024

Position: DevOps

Customer: OLE
Team size: 2
Tech stack: AWS (EKS, ECR, EFS, VPC), Terraform, Ansible, GitLab CI, (Argo CD), Helm, Prometheus, kube-state-metrics, Grafana, Loki, Fluent Bit, ingress-nginx, cert-manager, Docker, Python.

- **Automated provisioning** of **EKS** and core infrastructure with **Terraform/Ansible**, enabling on-demand cluster creation and **elastic scaling**.
- Expanded pod IP capacity by designing **secondary CIDR blocks** at the **VPC/cluster** layers to remove **IP constraints** and support growth.
- Led **GitLab CI/CD** implementation for **build/test/security scan/deploy** to **EKS** across **development, staging, and production**.
- Standardized deployments with reusable **Helm charts** stored in an internal **chart repository**, improving **consistency** and **collaboration**.
- Automated container **builds** and **image governance**; published **signed, versioned images** to **AWS ECR** with **retention/scan policies**.
- Provided **persistent storage** via **AWS EFS** and Kubernetes **PVCs**; **migrated stateful services** to the platform.

- Established **observability** with **Prometheus** and **kube-state-metrics** feeding **Grafana dashboards** for cluster, workload, and application health.
- Implemented **centralized logging** using **Fluent Bit** and **Loki** with **Grafana** for **search, exploration, and correlation**.
- Secured ingress with **ingress-nginx** and **cert-manager**; automated **certificate issuance/renewal** and enforced **HTTPS**.
- Designed **release strategies** (**blue-green/canary**) using **Helm values** and environment overlays; documented **runbooks** for **promotion** and **rollback**.
- Introduced **GitOps with Argo CD** (app-of-apps, environment overlays for **dev/stage/prod**) to codify desired state and streamline promotions.
- Operated the platform end-to-end: routine **k8s upgrades**, **policy/security hardening**, **capacity planning**, and best-practice guidance for application teams.

VPBank Call Center Apr 2020 - May 2022

Position: Administrator/Operator

Customer: VPBank
Team size: 5
Tech stack: Linux server, OpenSips, RTPProxy, Asterisk, PostgreSQL, Prometheus, Grafana, Homer, Heplify, Python, Jenkins, Git, Docker-compose, Ansible, SIP packets analysis.

- Provisioning an SIP proxy enhanced the network connection for QOS. Upgraded Opensips to integrated custom Prometheus metrics that give better monitoring experience and fast resolution of on-going issues as soon as possible.
- Customize the Grafana dashboard and provide it to the client for proactive monitoring.
- Technical support to analyze SIP packets for checking QOS.
- Configuration management with Git and Jenkins.

FE Credit Call Center June 2019 - May 2022

Position: Administrator/Operator

Customer: FE Credit
Team size: 4
Tech stack: Linux server, OpenSips, RTPProxy, Asterisk, PostgreSQL, Prometheus, Grafana, Homer, Heplify, Python, Jenkins, Git, Docker-compose, Ansible.

- Maintained **Asterisk SIP proxy** and configuration management with Git and Jenkins.
- I researched and replaced Asterisk with **OpenSips**, achieving better performance by 40% and optimizing resource usage.
- Collaborated with the developer team to optimize log management.
- **Containerized** components of the SIP proxy structure reduce deployment time by automatically provisioning processes using **Ansible** and **Docker-compose**.
- Implemented **Prometheus** and **Grafana** to monitor SIP proxy quality and server resources. Reduced impact for call quality on the SIP service.
- The integrated **Flask API** involved custom requests on the SIP proxy. Resulted in controlling SIP flow and customizing the logs in the database.

FIND ME ON

Github: https://github.com/mach1el
LinkedIn: https://www.linkedin.com/in/mich43l
Hackthebox: https://app.hackthebox.com/profile/83903
Hackerrank: https://www.hackerrank.com/profile/mich43l

INTERESTS

- Playing video game
- Doing workout
- Enjoy the music

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